Aufgabe 1: Addition im Binärsystem

Schriftliche Addition im Binärsystem funktioniert analog zur schriftlichen Addition im Dezimalsystem. Berechnen Sie das Ergebnis folgender Additionen: Bearbeiten Sie beide Additionen direkt im Aufgabenblatt.

1.)

	0	1	0	0	0	1	0	0	0
+	0	1	1	0	1	1	1	0	0
	1	0	1	1	0	0	1	0	0

Simulator ↗

2.)

	0	1	0	1	0	1	0	0	0
+	0	0	1	0	0	0	0	0	0
	0	1	1	1	0	1	0	0	0

Simulator ↗

3.)

	0	0	1	0	0	1	0	0	1
+	0	0	0	1	0	1	1	0	1
	0	0	1	1	1	0	1	1	0

Simulator ↗

4.)

	0	0	1	0	1	0	0	0	0
+	0	1	1	0	0	1	1	1	1
	1	0	0	0	1	1	1	1	1

Simulator ↗

5.)

	0	0	1	0	0	1	1	1	1
+	0	1	0	1	0	1	1	1	1
	0	1	1	1	1	1	1	1	0

Simulator ↗

6.)

	0	1	1	0	1	0	1	1	0
+	0	0	0	0	1	0	1	1	1
	0	1	1	1	0	1	1	0	1

Simulator ↗

7.)

	0	0	0	1	0	1	0	0	1
+	0	1	0	1	0	0	0	1	0
	0	1	1	0	0	1	0	1	1

Simulator ↗

8.)

	0	1	0	0	0	1	0	1	1
+	0	0	0	1	1	1	1	1	0
	0	1	1	0	0	1	0	0	1

Simulator ↗

9.)

	0	1	0	1	0	1	1	0	0
+	0	1	0	0	1	1	0	1	0
	1	0	1	0	0	0	1	1	0

Simulator ↗

10.)

	0	0	1	1	0	0	0	0	0
+	0	0	0	1	1	1	1	1	0
	0	1	0	0	1	1	1	1	0

Simulator ↗

11.)

	0	0	1	1	0	0	0	0	1
+	0	1	0	1	1	0	1	0	1
	1	0	0	0	1	0	1	1	0

Simulator ↗

12.)

	0	1	0	1	1	0	0	0	1
+	0	1	1	0	0	0	1	0	0
	1	0	1	1	1	0	1	0	1

Simulator ↗

13.)

	0	1	1	0	0	1	1	0	0
+	0	0	1	1	0	1	0	1	0
	1	0	0	1	1	0	1	1	0

Simulator

14.)

	0	1	1	1	0	1	0	1	0
+	0	1	0	0	0	1	0	0	1
	1	0	1	1	1	0	0	1	1

Simulator

15.)

	0	1	0	0	1	0	0	1	0
+	0	1	0	1	0	0	1	1	1
	1	0	0	1	1	1	0	0	1

Simulator

16.)

	0	0	0	0	0	1	1	1	1
+	0	0	0	1	0	1	1	1	1
	0	0	0	1	1	1	1	1	0

Simulator

17.)

	0	1	1	0	1	1	0	1	1
+	0	1	1	1	1	0	0	0	0
	1	1	1	0	0	1	0	1	1

Simulator

18.)

	0	1	1	1	0	1	1	0	1
+	0	1	0	0	0	0	1	0	0
	1	0	1	1	1	0	0	0	1

Simulator

19.)

	0	1	0	0	0	1	1	0	0
+	0	1	1	0	1	0	1	1	1
	1	0	1	1	0	0	0	1	1

Simulator

20.)

	0	0	1	1	0	1	1	0	0
+	0	0	0	1	1	0	1	0	1
	0	1	0	1	0	0	0	0	1

Simulator

21.)

	0	1	1	1	0	0	1	1	0
+	0	1	1	0	0	1	1	1	1
	1	1	0	1	1	0	1	0	1

Simulator

22.)

	0	1	1	1	0	1	1	1	1
+	0	0	0	1	0	1	1	0	1
	1	0	0	0	1	1	1	0	0

Simulator

23.)

	0	1	1	0	0	1	1	0	1
+	0	0	0	0	1	1	0	0	1
	0	1	1	1	0	0	1	1	0

Simulator

24.)

	0	0	1	1	0	1	0	0	0
+	0	1	1	0	0	1	0	1	0
	1	0	0	1	1	0	0	1	0

Simulator

25.)

	0	0	1	1	0	0	0	1	1
+	0	1	0	1	0	1	0	1	1
	1	0	0	0	0	1	1	1	0

Simulator

26.)

	0	1	1	1	0	1	1	1	0
+	0	0	1	1	0	0	1	1	0
	1	0	1	0	1	0	1	0	0

Simulator

27.)

$$\begin{array}{r} 0\,0\,0\,1\,1\,0\,1\,0\,1 \\ +\;0\,0\,0\,1\,1\,0\,0\,0\,1 \\ \hline 0\,0\,1\,1\,0\,0\,1\,1\,0 \end{array}$$

Simulator

28.)

$$\begin{array}{r} 0\,0\,0\,0\,1\,0\,0\,1\,1 \\ +\;0\,0\,0\,0\,1\,0\,1\,1\,1 \\ \hline 0\,0\,0\,1\,0\,1\,0\,1\,0 \end{array}$$

Simulator

29.)

$$\begin{array}{r} 0\,1\,0\,1\,1\,1\,0\,0\,0 \\ +\;0\,1\,1\,0\,1\,1\,1\,0\,1 \\ \hline 1\,1\,0\,0\,1\,0\,1\,0\,1 \end{array}$$

Simulator

30.)

$$\begin{array}{r} 0\,1\,0\,1\,1\,1\,0\,0\,1 \\ +\;0\,1\,1\,0\,1\,1\,1\,0\,1 \\ \hline 1\,1\,0\,0\,1\,0\,1\,1\,0 \end{array}$$

Simulator

31.)

$$\begin{array}{r} 0\,1\,1\,1\,0\,1\,0\,0\,1 \\ +\;0\,0\,1\,1\,1\,0\,1\,0\,0 \\ \hline 1\,0\,1\,0\,1\,1\,1\,0\,1 \end{array}$$

Simulator

32.)

$$\begin{array}{r} 0\,1\,0\,0\,0\,1\,1\,0\,0 \\ +\;0\,0\,0\,1\,1\,1\,1\,1\,1 \\ \hline 0\,1\,1\,0\,0\,1\,0\,1\,1 \end{array}$$

Simulator

33.)

$$\begin{array}{r} 0\,0\,0\,1\,1\,1\,1\,1\,1 \\ +\;0\,1\,1\,1\,1\,1\,1\,1\,1 \\ \hline 1\,0\,0\,1\,1\,1\,1\,1\,0 \end{array}$$

Simulator

34.)

$$\begin{array}{r} 0\,0\,0\,0\,0\,0\,0\,1\,0 \\ +\;0\,0\,0\,1\,1\,1\,0\,1\,0 \\ \hline 0\,0\,0\,1\,1\,1\,1\,0\,0 \end{array}$$

Simulator

35.)

$$\begin{array}{r} 0\,0\,1\,1\,0\,0\,0\,1\,1 \\ +\;0\,0\,0\,1\,1\,0\,0\,0\,0 \\ \hline 0\,1\,0\,0\,1\,0\,0\,1\,1 \end{array}$$

Simulator

36.)

$$\begin{array}{r} 0\,1\,1\,1\,0\,0\,1\,0\,1 \\ +\;0\,1\,1\,0\,1\,0\,0\,0\,1 \\ \hline 1\,1\,0\,1\,1\,0\,1\,1\,0 \end{array}$$

Simulator

37.)

$$\begin{array}{r} 0\,0\,0\,0\,0\,1\,0\,0\,0 \\ +\;0\,0\,0\,0\,0\,0\,1\,0\,0 \\ \hline 0\,0\,0\,0\,0\,1\,1\,0\,0 \end{array}$$

Simulator

38.)

$$\begin{array}{r} 0\,0\,1\,1\,1\,1\,1\,1\,0 \\ +\;0\,1\,0\,1\,0\,0\,1\,1\,0 \\ \hline 1\,0\,0\,1\,0\,0\,1\,0\,0 \end{array}$$

Simulator

39.)

$$\begin{array}{r} 0\,1\,1\,1\,0\,1\,1\,1\,1 \\ +\;0\,0\,1\,0\,0\,1\,0\,1\,0 \\ \hline 1\,0\,0\,1\,1\,1\,0\,0\,1 \end{array}$$

Simulator

40.)

$$\begin{array}{r} 0\,0\,0\,1\,0\,1\,0\,0\,0 \\ +\;0\,0\,1\,1\,0\,1\,0\,0\,1 \\ \hline 0\,1\,0\,0\,1\,0\,0\,0\,1 \end{array}$$

Simulator

41.)

$$\begin{array}{r} 0\,1\,0\,1\,1\,1\,0\,1\,0 \\ +\;0\,0\,0\,0\,0\,1\,0\,1\,0 \\ \hline 0\,1\,1\,0\,0\,0\,1\,0\,0 \end{array}$$

Simulator ↗

42.)

$$\begin{array}{r} 0\,0\,1\,0\,0\,0\,0\,0\,0 \\ +\;0\,0\,0\,1\,1\,0\,0\,0\,1 \\ \hline 0\,0\,1\,1\,1\,0\,0\,0\,1 \end{array}$$

Simulator ↗

43.)

$$\begin{array}{r} 0\,1\,0\,1\,0\,1\,0\,0\,1 \\ +\;0\,0\,0\,1\,0\,1\,1\,0\,0 \\ \hline 0\,1\,1\,0\,1\,0\,1\,0\,1 \end{array}$$

Simulator ↗

44.)

$$\begin{array}{r} 0\,1\,0\,0\,0\,1\,1\,0\,0 \\ +\;0\,0\,1\,1\,1\,0\,0\,1\,0 \\ \hline 0\,1\,1\,1\,1\,1\,1\,1\,0 \end{array}$$

Simulator ↗

45.)

$$\begin{array}{r} 0\,1\,0\,1\,0\,0\,1\,1\,1 \\ +\;0\,0\,0\,0\,0\,1\,0\,0\,0 \\ \hline 0\,1\,0\,1\,0\,1\,1\,1\,1 \end{array}$$

Simulator ↗

46.)

$$\begin{array}{r} 0\,0\,0\,0\,0\,1\,1\,0\,1 \\ +\;0\,0\,1\,1\,1\,1\,0\,1\,0 \\ \hline 0\,1\,0\,0\,0\,0\,1\,1\,1 \end{array}$$

Simulator ↗

47.)

$$\begin{array}{r} 0\,1\,0\,0\,0\,0\,1\,0\,1 \\ +\;0\,0\,1\,1\,0\,0\,1\,0\,1 \\ \hline 0\,1\,1\,1\,0\,1\,0\,1\,0 \end{array}$$

Simulator ↗

48.)

$$\begin{array}{r} 0\,0\,1\,0\,0\,1\,0\,0\,0 \\ +\;0\,1\,1\,1\,1\,0\,1\,0\,0 \\ \hline 1\,0\,0\,1\,1\,1\,1\,0\,0 \end{array}$$

Simulator ↗

49.)

$$\begin{array}{r} 0\,1\,0\,0\,0\,1\,1\,0\,0 \\ +\;0\,0\,1\,0\,0\,0\,1\,1\,1 \\ \hline 0\,1\,1\,0\,1\,0\,0\,1\,1 \end{array}$$

Simulator ↗

50.)

$$\begin{array}{r} 0\,1\,0\,0\,1\,1\,0\,1\,0 \\ +\;0\,1\,0\,0\,0\,0\,0\,0\,0 \\ \hline 1\,0\,0\,0\,1\,1\,0\,1\,0 \end{array}$$

Simulator ↗